COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 - Data Structures & Algorithms	**SECTION:**	CPE21S2
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ACTIVITY NO 6.1 **Assignment 6.1 Linear and Binary Search**

1. PROCEDURE OUTPUT

1. What is a search tree in data structures?
A search tree is a data structure used to store and organize data in a way that makes searching, inserting, and deleting elements easier. A common example is a Binary Search Tree (BST), where smaller values are placed on the left side, and larger values are placed on the right side of a node.

2. What are the Different types of search algorithms in data structures? Differentiate each type of search.
The two common types of search algorithms are linear search and binary search. Linear search checks each element one by one until the desired element is found, while binary search checks the middle element and repeatedly divides the search area into two. Linear search can be used on both sorted and unsorted data, while binary search requires the data to be sorted.

3. What operations/implementations can be performed using binary and linear search operations?
Linear and binary search can be used to find a specific element, determine whether an element exists, and find the position or index of an element in a collection of data. Linear search can also be used to search unsorted data, while binary search is commonly used for efficiently searching sorted data.

4. What are the advantages in using a binary search tree as a data structure?
A Binary Search Tree allows data to be searched, inserted, and deleted efficiently, especially when the tree is balanced. It also keeps data organized in a sorted manner, and an in-order traversal can display the elements in ascending order. A BST can also grow dynamically as new elements are added.

5. Give an example program using binary search and Linear search.

```cpp
#include <iostream>

// Linear Search
int LinearSearch(int arr[], int size, int target) {
    for (int i = 0; i < size; i++) {
        if (arr[i] == target) {
            std::cout << "Target found!";
            return i;
        }
        else
            if (i == size - 1) {
                std::cout << "Target not found in the list.";
        }
    }
    return -1;
}

// Binary Search
int BinarySearch(int arr[], int size, int target) {
    int low = 0, high = size - 1;
    int f = 0, i;

    while (low <= high) {
        int mid = (low + high) / 2;

        if (arr[mid] == target) {
            std::cout << "Target found in the List!";
            return mid;
        }
        else if (arr[mid] > target) {
            high = mid - 1;
        }
        else {
            low = mid + 1;
        }
    }


    std::cout << "Target not found in the list.";
    return -1;
}
```

```cpp
// main
int main() {
    int list1[5] = {7, 9, 12, 15, 38};
    int size1 = 5;

    int list2[5] = {2, 5, 6, 12, 38};
    int size2 = 5;

    int number;

    std::cout << "Please Enter a Number to Search in the list: ";
    std::cin >> number;

    int linearResult = LinearSearch(list1, size1, number);
    if (linearResult != 1)
        std::cout << "\nLinear Search has determined its in: " << linearResult << std::endl;

    int binaryResult = BinarySearch(list2, size2, number);
    if (binaryResult != 1)
        std::cout << "\nBinary Search has determined its in: " << binaryResult << std::endl;

    return 0;
}
```

```
Please Enter a Number to Search in the list: 0
Target not found in the list.
Linear Search has determined its in: -1
Target not found in the list.
Binary Search has determined its in: -1
```

```
Please Enter a Number to Search in the list: 38
Target found!
Linear Search has determined its in: 4
Target found in the List!
Binary Search has determined its in: 4
```

When the user enters 0, both linear and binary search return -1, meaning that the target number was not found in either list. When the user enters 12, linear search finds it at index 2 in the first list, while binary search finds it at index 3 in the second list.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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